CLASS 12 MATHEMATICS

130 Essential Questions (10 Per Chapter) for Board Exams

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Chapter 01: Relations and Functions

1. Is the function $f: Z \to Z$ such that $f(x) = x^2 + x$ injective, surjective or bijective?

2. If $f: R \to R$ be the function defined by $f(x) = 4x^3 + 7$, show that $f(x)$ is a bijection.

3. Check whether the relation R defined on the set A = {1, 2, 3, 4, 5, 6} as R = {(a, b) : b = a + 1} is reflexive, symmetric or transitive.

4. Show that the relation R in the set $A = \{x \in Z : 0 \leq x \leq 12\}$ given by R = {(a, b) : |a - b| is divisible by 4} is an equivalence relation.

5. Show that the relation S defined on set $N \times N$ by (a, b)S(c, d) if a + d = b + c is an equivalence relation.

6. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by (a, b)R(c, d) if ad(b + c) = bc(a + d). Show that R is an equivalence relation.

7. Prove that the relation R in the set A = {1, 2, 3, 4, 5} given by R = {(a, b) : |a - b| is even} is an equivalence relation.

8. Show that the function f in A = R - {2/3} defined as f(x) = (4x+3)/(6x-4) is one-one and onto.

9. Write the domain of the relation R defined on the set Z of integers as follows: $(a, b) \in R \leftrightarrow a^2 + b^2 = 25$.

10. Show that f: R+ → R+ defined by f(x) = 1/2x is bijective, where R+ is the set of all non-zero positive real numbers.

Chapter 02: Inverse Trigonometric Functions

1. Find the number of triplets (x, y, z) satisfying the equation $\sin^{1}x + \sin^{1}y + \sin^{1}z = 3\pi/2$.

2. Solve for x: $\cos(\tan^{1}x) = \sin(\cot^{1} 3/4)$.

3. Find the value of $\sin^{1}(\cos(43\pi/5))$.

4. Write the principal value of $\cos^{1}(1/2) + 2\sin^{1}(1/2)$.

5. Evaluate: $\tan\{2\tan^{1}(1/5) + \pi/4\}$.

6. Find the domain of the function $f(x) = \sin^{1}(x^2 - 4)$. Also, find its range.

7. Prove that: $9\pi/8 - 9/4\sin^{1}(1/3) = 9/4\sin^{1}(2\sqrt{2} / 3)$.

8. Prove that: $\tan^{1}[(\sqrt{(1+x)} + \sqrt{(1-x)}) / (\sqrt{(1+x)} - \sqrt{(1-x)})] = \pi/4 - 1/2\cos^{1}x$.

9. Find the number of solutions of the equation $2\cos^{1}x + \sin^{1}x = 11\pi/6$.

10. Find the domain of the function defined by $f(x) = \sin^{1} \sqrt{(x-1)}$.

Chapter 03: Matrices

1. Simplify: $(A - I)^3 + (A + I)^3 - 7A$, if $A^2 = I$.

2. If the matrix A = [0 1 -2; -1 0 3; x -3 0] is skew-symmetric, find x.

3. Find X and Y if X + Y = [5 2; 0 9] and X - Y = [3 6; 0 -1].

4. Express the matrix A = [2 4 -6; 7 3 5; 1 -2 4] as the sum of a symmetric and a skew-symmetric matrix.

5. If A = [3 1; -1 2], show that $A^2 - 5A + 7I = 0$. Hence find A^{1}.

6. For the matrix A = [1 1 1; 1 1 1; 1 1 1], show that $A^n = [3^{n1}]$ (all elements) by Mathematical Induction.

7. Find the value of k if A = [1 0; -1 7] and $A^2 = 8A + kI$.

8. Given A = [2 -3; -4 7], compute A^{1} and show that $2A^{1} = 9I - A$.

9. If A = [0 -tan(x/2); tan(x/2) 0] and I is the identity matrix of order 2, show that I + A = (I - A) [cos x - sin x; sin x cos x].

10. Find a matrix A such that 2A - 3B + 5C = 0, where B and C are given matrices.

Chapter 04: Determinants

1. If A = [1 2; 3 4], find adj A.

2. If A is a square matrix satisfying A'A = I, write the value of |A|.

3. If A = [4 6; 7 5], then what is the value of A · (adj A)?

4. For what value of x is the matrix [3-2x x+1; 2 4] singular?

5. If A, B are square matrices of the same order, prove that adj(AB) = (adj B)(adj A).

6. Solve the system of equations using matrix method: $2x + y - 3z = 13$, $3x + 2y + z = 4$, $x + 2y - z = 8$.

7. If A = [2 3; 1 -4] and B = [1 -2; -1 3], verify that $(AB)^{1} = B^{1}A^{1}$.

8. Given A = [2 -3 5; 3 2 -4; 1 1 -2], find A^1. Using A^1 solve the system of equations.

9. If A = [2 -1; -1 2] and I is the identity matrix of order 2, show that $A^2 = 4A - 3I$. Hence find A^1.

10. If A = [cos α -sin α 0; sin α cos α 0; 0 0 1], find adj A and verify that A(adj A) = (adj A)A = $|A|I_3$.

Chapter 05: Continuity and Differentiability

1. Find the value of a, if the function f(x) defined by f(x) = {2x-1, x<2; a, x=2; x+1, x>2} is continuous at x=2.

2. Show that the function f(x) = |x-3| is continuous but not differentiable at x=3.

3. Find the value of k so that the function f(x) = {k cos x / (π-2x), x ≠ π/2; 3, x = π/2} is continuous at x = π/2.

4. Find the values of p and q for which f(x) = {$(1-\sin^3x)/3\cos^2x$, x<π/2; p, x=π/2; $q(1-\sin x)/(\pi-2x)^2$, x>π/2} is continuous at x = π/2.

5. If y = e^(x cos x) + (cos x)^x, then find dy/dx.

6. If x^y = e^(x-y), then show that dy/dx = log x / $(1+\log x)^2$.

7. Differentiate $\tan^1$ [(√$(1+x^2)$-1)/x] with respect to $\tan^1 x$.

8. Find the relationship between a and b so that the function f defined by f(x) = {ax+1, x ≤ 3; bx+3, x > 3} is continuous at x=3.

9. If y = Ae^(mx) + Be^(nx), show that d^2y/dx^2 - (m+n)dy/dx + mny = 0.

10. If x = a(cos t + log tan t/2), y = a sin t, find d^2y/dt^2 and d^2y/dx^2.

Chapter 06: Application of Derivatives

1. Find the intervals in which the function $f(x) = x^4/4 - x^3 - 5x^2 + 24x + 12$ is (i) strictly increasing, (ii) strictly decreasing.

2. An open tank with a square base and vertical sides is to be constructed from a metal sheet. Show that the cost of material will be least when depth of the tank is half of its width.

3. Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is 2R/√3. Also find the max volume.

4. A ladder 13 m long is leaning against a vertical wall. The bottom is dragged away at 2 cm/sec. How fast is the height decreasing when the foot is 5 m away?

5. Show that the semi-vertical angle of a cone of maximum volume and given slant height is $\cos^1$(1/√3).

6. Prove that the least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is $6\sqrt{3}r$.

7. Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $4r/3$.

8. Find the maximum and minimum values of $f(x) = -|x-1| + 5$ for all $x \in R$.

9. The side of an equilateral triangle is increasing at the rate of 2 cm/s. At what rate is its area increasing when the side is 20 cm?

10. Prove that the function f defined by $f(x) = x^2 - x + 1$ is neither increasing nor decreasing in $(-1, 1)$.

Chapter 07: Integrals

1. Evaluate: $\int (x-4)e^x / (x-2)^3 \, dx$.

2. Evaluate: $\int dx / \sqrt{(5-4x-2x^2)}$.

3. Evaluate: $\int \sin 2x / [(1+\sin x)(2+\sin x)] \, dx$.

4. Evaluate: $\int (\cos 2x - \cos 2\alpha) / (\cos x - \cos \alpha) \, dx$.

5. Evaluate: $\int$ from 0 to $\pi/2$ $[\sin x / (\sin x + \cos x)] \, dx$.

6. Evaluate: $\int e^x [(1+\sin x)/(1+\cos x)] \, dx$.

7. Evaluate: $\int (5x+3) / \sqrt{(x^2+4x+10)} \, dx$.

8. Evaluate: $\int (\sqrt{\cot x} + \sqrt{\tan x}) \, dx$.

9. Find: $\int (x+2) / [(x^2+3x+3)\sqrt{(x+1)}] \, dx$.

10. Evaluate: $\int$ from 0 to $\pi/4$ $[\log(1+\tan x)] \, dx$.

Chapter 08: Application of Integrals

1. Using integration, find the area of the region $\{(x, y) : x^2 + y^2 \le 2ax, y^2 \ge ax, x, y \ge 0\}$.

2. Using integration, find the area of the region bounded by the curve $x^2 = 4y$ and the line $x = 4y - 2$.

3. Using integration, find the area of the region bounded by the triangle whose vertices are $(-1, 2)$, $(1, 5)$ and $(3, 4)$.

4. Find the area of the smaller region bounded by the ellipse $x^2/9 + y^2/4 = 1$ and the line $x/3 + y/2 = 1$.

5. Find the area of the region $\{(x, y) : x^2 + y^2 \le 4, x + y \ge 2\}$.

6. Find the area bounded by the curve $y = x|x|$, x-axis and the lines $x = -3$ and $x = 3$.

7. Find the area of the region bounded by the curve $4x^2 + y^2 = 36$ using integration.

8. Using integration, find the area of the region enclosed between two circles $x^2 + y^2 = 4$ and $(x-2)^2 + y^2 = 4$.

9. Find the area of the region in the first quadrant enclosed by the x-axis, the line $y=x$ and the circle $x^2 + y^2 = 32$.

10. Find the area bounded by the circle $x^2 + y^2 = 16$ and the line $\sqrt{3}y = x$ in the first quadrant.

Chapter 09: Differential Equations

1. Solve the differential equation: $(1+x^2)dy + 2xy\, dx = \cot x\, dx$.

2. Solve the differential equation: $(x+1)dy/dx = 2e^{(\ddot{y})} - 1$; $y(0) = 0$.

3. Find the general solution: $e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$.

4. Find the particular solution: $dy/dx = 1 + x^2 + y^2 + x^2y^2$, given $y=1$ when $x=0$.

5. Find the particular solution: $2xy + y^2 - 2x^2\, dy/dx = 0$; $y=2$ when $x=1$.

6. Solve: $x \cos(y/x)\, dy/dx = y \cos(y/x) + x$.

7. Solve: $(1+y^2) + (x - e^{(\tan^1 y)})\, dy/dx = 0$.

8. Show that $2ye^{(x/y)}\, dx + (y - 2xe^{(x/y)})\, dy = 0$ is homogeneous. Find the particular solution given $x=0$ when $y=1$.

9. Find the particular solution: $dy/dx - 3y \cot x = \sin 2x$, given $y=2$ when $x=\pi/2$.

10. Solve: $(\tan^1 y - x)dy = (1+y^2)dx$.

Chapter 10: Vector Algebra

1. If $a = 2i + 4j - 5k$ and $b = i + 2j + 3k$, find the unit vector in the direction of $a + b$.

2. Using vectors, find the area of triangle ABC with vertices A(1, 2, 3), B(2, -1, 4) and C(4, 5, -1).

3. Find the projection of vector $(i - j)$ on $(i + j)$.

4. If $|a + b| = |a|$, prove that $2a + b$ is perpendicular to b.

5. Find vector p perpendicular to both $\alpha = 4i + 5j - k$ and $\beta = i - 4j + 5k$, given $p \cdot q = 21$ where $q = 3i + j - k$.

6. If $a = i - j + 7k$ and $b = 5i - j + \lambda k$, find λ so that $a + b$ and $a - b$ are perpendicular.

7. Find the area of a parallelogram whose adjacent sides are a = i - j + 3k and b = 2i - 7j + k.

8. If |a|=3, |b|=4, |c|=5 and each is perpendicular to the sum of the other two, find |a + b + c|.

9. Find a unit vector perpendicular to both (3a + 2b) and (3a - 2b), where a = i+j+k and b = i+2j+3k.

10. If a + b + c = 0 and |a|=3, |b|=5, |c|=7. Find the angle between a and b.

Chapter 11: Three-Dimensional Geometry

1. Find the shortest distance between [(x-1)/2 = (y-2)/3 = (z+1)/4] and [(x+2)/-1 = (y-3)/2 = z/3].

2. Find the vector equation of the line passing through (2, 3, -5) and making equal angles with co-ordinate axes.

3. Find equation of line bisecting segment A(2, 3, 4) and B(4, 5, 8) and perpendicular to [(x-8)/3 = (y+19)/-16 = (z-10)/7] and [(x-15)/3 = (y-29)/8 = (z-5)/-5].

4. Find equation of line through (1, 2, -4) and perp to r = (8i-19j+10k)+λ(3i-16j+7k) and r = (15i-29j+5k)+μ(3i+8j-5k).

5. Find foot of perpendicular, length and image of P(5, 4, 2) on r = (-i+3j+k)+λ(2i+3j-k).

6. Show that lines [(x+1)/3 = (y+3)/5 = (z+5)/7] and [(x-2)/1 = (y-4)/3 = (z-6)/5] intersect. Find point of intersection.

7. Find vector/cartesian equations of line through (1, 2, -4) perp to [(x-8)/3 = (y+19)/-16 = (z-10)/7] and [(x-15)/3 = (y-29)/8 = (z-5)/-5].

8. Find p so that [(1-x)/3 = (7y-14)/2p = (z-3)/2] and [(7-7x)/3p = (y-5)/1 = (6-z)/5] are perpendicular.

9. Find direction cosines of [(x+2)/2 = (2y-7)/6 = (5-z)/6]. Find vector eq of line through A(-1, 2, 3) parallel to it.

10. If a line makes 60° and 45° with x and z axes, find angle it makes with positive y-axis.

Chapter 12: Linear Programming

1. Maximize z = 2x + 5y, subject to: 2x + 4y ≤ 8, 3x + y ≤ 6, x + y ≤ 4, x ≥ 0, y ≥ 0.

2. Maximize Z = 7x + 10y, subject to: 4x + 6y ≤ 240, 6x + 3y ≤ 240, x ≥ 10, x, y ≥ 0.

3. Minimize Z = 5x + 7y, subject to: 2x + y ≥ 8, x + 2y ≥ 10, x, y ≥ 0.

4. Minimize Z = 5x + 10y, subject to: x + 2y ≤ 120, x + y ≥ 60, x - 2y ≥ 0, x, y ≥ 0.

5. Maximize P = 70x + 40y, subject to: 3x + 2y ≤ 9, 3x + y ≤ 9, x, y ≥ 0.

6. Maximize $Z = x + 2y$, subject to: $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x, y \geq 0$.

7. Find max of $Z = X + Y$, subject to: $y \geq |x| - 1$, $y \leq 1 - |x|$, $x, y \geq 0$.

8. Maximize $z = 2x + 5y$, subject to: $2x + 4y \leq 8$, $3x + y \leq 6$, $x + y \leq 4$, $x, y \geq 0$.

9. Maximize $z = 8x + 9y$, subject to: $2x + 3y \leq 6$, $3x - 2y \leq 6$, $y \leq 1$, $x, y \geq 0$.

10. Minimize $Z = 30x + 20y$, subject to: $x + y \leq 8$, $x + 4y \geq 12$, $5x + 8y \geq 20$, $x, y \geq 0$.

Chapter 13: Probability

1. Purse A (3S, 6C coins), Purse B (4S, 3C). Draw a coin, find prob it is silver.

2. Die faces 1,2,3 red; 4,5,6 green. A: even, B: red. Are A and B independent?

3. A and B throw dice alternately. A wins on 7, B on 10. A starts. Prob B wins?

4. If $P(E)=7/13$, $P(F)=9/13$, $P(E \cap F)=4/13$. Evaluate $P(E/F)$.

5. Card lost from deck. Draw 1 from rest, it's a King. Find prob lost was a King.

6. Biased die (even twice as likely as odd). Tossed twice. Prob distribution of sixes.

7. Probs of A,B,C,D solving: 1/3, 1/4, 1/5, 2/3. Prob (i) solved, (ii) at most 1 sol.

8. A truths 75%, B 90%. What % cases do they contradict each other?

9. Bag has $(2n+1)$ coins. $(n-1)$ are double-headed, rest fair. Toss gives head ($p=31/42$). Find n.

10. Machines A(30%), B(50%), C(20%) make bolts. Defectives: 3%, 4%, 1%. Draw defective, prob Not from B.